

Problem A

Lynn's Carrot Worker Problem

Time limit: 3 seconds

Memory limit: 1024 megabytes

Problem Description

Lynn is a computer science freshman. Besides happily coding, one of her favorite hobbies is a carrot-farming simulation game, where she watches her cute carrots run around, fish, raise chickens, and process orders by commanding them to prepare ingredients.

One day, while she and Wenci were at the university cafeteria, playing the game and grinding LeetCode.

Lynn: "Wait... no. That's just wrong."

Wenci: (Startled) "Huh? What's wrong?"

Lynn: "Think about it! I have all these carrot workers, and they have to finish tasks in order. But as soon as one worker finishes his block, he just bails and goes to sleep!"

Wenci: "Yeah... so? What's the problem?"

Lynn: "So, if I have N tasks and I have to split them among K workers... isn't one of them going to be totally overworked? The one stuck with the longest continuous block of work? I need to write a program to minimize that max workload."

Wenci: "Whoa, chill out. You're thinking about Leetcode again, aren't you?"

But Lynn was already too excited, pulling out her laptop to solve this problem.

This problem is to help Lynn find the optimal solution. You are given N tasks that must be completed in sequence, and you have K carrot workers. The tasks must be partitioned among the K workers in consecutive blocks:

- Worker 1 will be assigned a consecutive block of tasks starting from task 1.
- Worker 2 will be assigned the next consecutive block of tasks.
- ...
- Worker K will be assigned the final consecutive block of tasks, finishing at task N .

Note that, workers are used in order. Once a worker finishes their assigned block, they are done and cannot take later tasks.

Your goal is to find a partitioning strategy that minimizes the maximum total time assigned to any single worker – i.e., minimize the working time of the "most tired carrot".

Input Format

Your program is to read from standard input. The input consists of multiple test cases until EOF. Each test case consists of two lines. The first line contains two integers N and K , representing the number of tasks and the number of carrot workers. The second line contains N integers representing the time required for each of the N tasks in order.

Output Format

For each test case, print exactly one line containing a single integer representing the minimized maximum working time for the most tired worker.

Technical Specification

- $1 \leq K \leq N \leq 2 \times 10^5$
- $0 \leq \text{time for each single task} \leq 10^4$

Sample Input 1

```
5 3
10 2 8 4 6
3 2
100 100 100
```

Sample Output 1

```
10
200
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